

SE3080 — Software Project Management

Assignment 01: Mid-Project Agile Review

Elderly Companionship & Micro-Volunteering Platform

Group ID: Group_018 | Opened: 28 Jul 2026, 12:00 AM — Due: 25 Aug 2026, 1:00 PM

Registration No.	Name	Scrum Role
IT23600416	Wijekoon W M V M	Product Owner + Development Team
IT23543300	Ranketh K A D	Scrum Master + Development Team
IT23534254	Pathirana P P D S S	Development Team
IT23670334	Perera K K L A	Development Team

Repository: https://github.com/dilmithp/SE3080-Group18_Assignment01

1. Project Selection and Justification

Group 018 is using the same product concept already developed for the UEE module, as directed: the Elderly Companionship & Micro-Volunteering Platform, a mobile app matching elderly residents who need occasional everyday help (grocery runs, walks, conversation, basic tech support) with local student/community volunteers. A user posts a need or an availability, the system matches by locality, skill and availability, a session is booked, and post-session feedback drives a trust score. This is a real, growing problem — elderly residents increasingly live alone with no structured, trust-verified way to get informal help — and existing volunteering apps are rarely built accessibility-first. The stakeholder voice is the elderly-resident and student-volunteer personas already validated in the parallel UEE (SE3050) module, used in every sprint review.

Guideline	How the project satisfies it
Real-world relevance	Genuine elderly-isolation / everyday-help need; persona-based stakeholder proxy reviews every sprint.
Agile suitability	Decomposes into 4 vertical epics, each independently demoable per sprint.
Scope manageability	Flutter + Firebase (managed backend, no infrastructure); scope fixed to 4 components across Sprint 0–2.
Team collaboration	Each of 4 members owns one component end-to-end; branch-per-feature + mandatory PR review.
Demo-ready increments	Every route/screen renders each sprint even while backend logic is still being wired up.
Tech flexibility	Flutter + Firebase chosen for velocity, not imposed by the brief.
Version control	Git repository maintained from Sprint 0 with a feature-branch + PR workflow.

2. Project Charter (Product Vision)

Vision: for elderly residents who need occasional help and companionship, and volunteers who want to give a few local hours, the platform matches the two by locality, skill and availability and builds a visible trust score from every completed session — designed accessibility-first from the ground up.

In scope (Sprint 0–2)	Out of scope for Assignment 01
Auth/RBAC/verification, profiles + accessibility framework, discovery/matching, session data model + calendar shell.	Sprint 3–4 polish (activity feed/streaks, release build); no payment processing; SMS gateway deferred.

Objectives: ship an accessible, navigable increment every sprint; keep each member's component independently testable and explainable; enforce a clean architecture boundary so features build in parallel without conflicts; evidence individual contribution via commit history, not just group demos.

Constraints: fixed 4-person team, Sprint 0–2 only for this assignment (28 Jul – 24 Aug 2026); accessibility (min font size, contrast ratio, tap-target size) is a hard requirement on every screen; Cloud Functions logic (matching, trust score, reminders) stubbed in Sprint 0/1, implemented in Sprint 2; no physical infrastructure or hardware.

Stakeholder	Role
Wijekoon (Product Owner)	Owns/prioritises the backlog; validates sprint review feedback.
Ranketh (Scrum Master)	Facilitates ceremonies; owns the board and burndown.
Pathirana, Perera (Dev Team)	Build and own their components end-to-end.
Elderly / volunteer personas	Stakeholder proxy — reviewed each sprint increment.
Module coordinator	Project sponsor for assessment purposes.

3. Initial Product Backlog

162 SP total across 4 epics (Fibonacci scale, MoSCoW priority): 68 SP in Sprint 1, 94 SP in Sprint 2 — Sprint 2 is heavier because it carries the algorithmic and Firebase-integration work. A further backlog (community feed/badges, escalation-flow hardening, SMS fallback, usability testing, release build) is groomed and reserved for Sprint 3–4, outside this assignment's scope.

Epic	ID	User story (condensed)	Pri	SP	Sprint
E1	US1.1	Register/login via email/phone OTP.	Must	8	S1
E1	US1.2	Role-based access via Firestore security rules.	Must	13	S1
E1	US1.3	Admin reviews/approves volunteer verification documents.	Must	8	S2
E1	US1.4	Trust score auto-updates from completed sessions/ratings.	Should	13	S2
E2	US2.1	Onboarding wizard + profile create/edit screens.	Must	8	S1
E2	US2.2	Accessibility widgets: text scaling, contrast, simplified nav.	Must	13	S1
E2	US2.3	Profile data (skills, availability, locality, photo) in Firestore/Storage.	Must	13	S2
E2	US2.4	Shared design system applied consistently across all 4 components.	Should	8	S2
E3	US3.1	Browse/search UI with category and distance filters.	Must	5	S1
E3	US3.2	Basic locality-based Firestore query behind search.	Must	8	S1
E3	US3.3	Matching Cloud Function: locality + skill + availability + trust.	Must	13	S2
E3	US3.4	Ranked results with match-reason explanation.	Should	13	S2
E4	US4.1	Session data model + availability calendar shell.	Must	5	S1
E4	US4.2	Session request creation with basic state transitions.	Must	8	S1
E4	US4.3	Full booking lifecycle with conflict handling	Must	13	S2

		and cancellation.			
E4	US4.4	Reminders (FCM) + post-session feedback loop.	Should	13	S2

Epics: E1 Identity/Access/Trust (Pathirana) · E2 Profiles/Accessibility (Perera) · E3 Matching/Discovery (Wijekoon) · E4 Scheduling/Feedback (Ranketh). SP reconciles to the Sprint 0 workload weighting: Pathirana 42, Perera 42, Wijekoon 39, Ranketh 39.

4. Sprint Backlogs and Plans (Sprint 0, 1 & 2)

Definition of Done: PR reviewed by ≥1 teammate (≥2 for core/); flutter analyze clean; unit tests added where the auth_trust pattern applies; UI meets the Sprint 0 accessibility standards; merged to dev only via a reviewed PR; docs/DATA_MODEL.md and firestore.rules updated in the same PR if the data model changed.

Sprint 0 — Setup (28 Jul – 3 Aug 2026)

Member	Tasks
Pathirana	Flutter project + repo structure; Firebase project (Auth/Firestore/Storage); GitHub branching, README, PR template.
Perera	Shared design system (Figma + Flutter theme/widgets); accessibility standards (font size, contrast, tap-target).
Wijekoon	Project Charter; Initial Product Backlog; confirmed stakeholder/customer proxy.
Ranketh	Sprint board + burndown tracking; Definition of Done; drafted Sprint 1 & 2 plans with PO.

Sprint 1 — 4-14 Aug 2026 — Goal: a verified user can register, build a profile and be discovered

Member	Tasks	SP
Pathirana	OTP registration/login; RBAC + Firestore security rules.	21
Perera	Profile data model + create/edit screens; accessibility widgets; onboarding wizard.	21
Wijekoon	Browse/search UI + filters; locality query; ran Sprint 1 review.	13
Ranketh	Facilitated ceremonies; burndown; session model + calendar shell; QA'd auth flows.	13

Sprint 2 — 15-24 Aug 2026 — Goal: a user can find a match, book a session and leave feedback

Member	Tasks	SP
Pathirana	Trust-score Cloud Function + badges; report/block + escalation flow; demo build/tagging.	21
Perera	High-fidelity matching/booking/feedback screens; design-system consistency; demo prep.	21
Wijekoon	Matching Cloud Function + ranked results; ran Sprint 2 review; re-prioritised backlog.	26
Ranketh	Full booking lifecycle + conflict handling; reminders + feedback loop; usability testing.	26

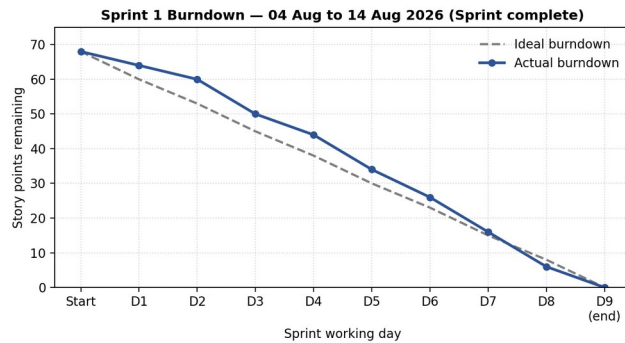


Fig 1 — Sprint 1 burndown (68 SP, delivered on schedule).

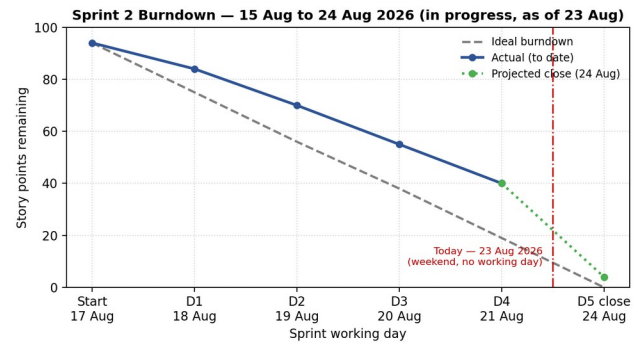


Fig 2 — Sprint 2 burndown, as of 23 Aug (~40 SP remaining vs ideal ~19; 24 Aug close projected).

5. Defined Scrum Roles and Responsibilities

Roles are fixed across Sprint 0–2. The PO and SM also contribute as developers, so all four members own a substantial, independently explainable component for the viva.

Role	Member	Responsibilities
Product Owner	Wijekoon (IT23600416)	Owns/prioritises backlog; presents at planning; validates stakeholder feedback at review.
Scrum Master	Ranketh (IT23543300)	Facilitates stand-ups/planning/retro; maintains board and burndown; removes blockers.
Development Team	All 4 (incl. PO & SM)	Estimate/commit at planning; implement owned components; review teammates' PRs.

6. Evidence of Scrum Events

Ceremony	Frequency	Lead
Sprint Planning	Start of sprint	Ranketh facilitates; Wijekoon presents backlog
Daily Stand-up	Daily, 10 min (Slack)	Ranketh
Sprint Review	End of sprint	Pathirana & Perera demo; Wijekoon validates
Retrospective	End of sprint	Ranketh
Backlog Refinement	Mid-sprint	Wijekoon

- Sprint 0 Planning (28 Jul): agreed the 4-component split, tech stack, and accessibility constraints.
- Sprint 1 Planning (4 Aug): pulled 68 SP; goal — a verified user can register, build a profile and be discovered.
- Sprint 1 Review (14 Aug): demoed OTP sign-in through saved profile + onboarding; persona feedback flagged small OTP-timer text, fixed under US2.4.
- Sprint 1 Retro (14 Aug): vertical split kept merge conflicts near zero; action item — add an accessibility check to the DoD review step (done for Sprint 2).
- Sprint 2 Planning (15 Aug): pulled 94 SP, front-loading the highest-risk Firebase-integration stories; goal — find a match, book a session, leave feedback.

Compiled from sprint-board activity, PR history and each member's ceremony notes; the team will verify specifics against their own memory before submission, per the CLEAR framework.

7. Technology Stack and Architecture

Layer	Technology	Why
Mobile front-end	Flutter (Dart)	One codebase, Android/iOS; strong accessibility widget support.
Backend / DB	Firebase Firestore	Real-time NoSQL — live session/booking

		status.
Auth	Firebase Authentication	Email/phone OTP — low-friction for elderly users.
Cloud logic	Cloud Functions	Matching, trust score, scheduled reminders.
Notifications / Storage	FCM / Firebase Storage	Push reminders; profile photos & verification docs.
State/DI/Router/DTOs	Riverpod, go_router, freezed, dartz	Explicit DI, declarative routing, functional error handling (Either).
Testing / VCS / PM	flutter_test+mocktail / Git+GitHub / Trello	Interface-level mocks; PR-reviewed feature branches; sprint board.

Feature-first Clean Architecture: presentation/domain/data per feature under a shared core/. Only data/ imports Firebase; domain/ has zero Firebase imports (PR-enforced). Repositories return Either<Failure, T> rather than throwing. New behaviour = new interface implementation (see MatchingStrategy) — Open/Closed in practice, not a branch in existing code.

elderly_companion — Feature-First Clean Architecture

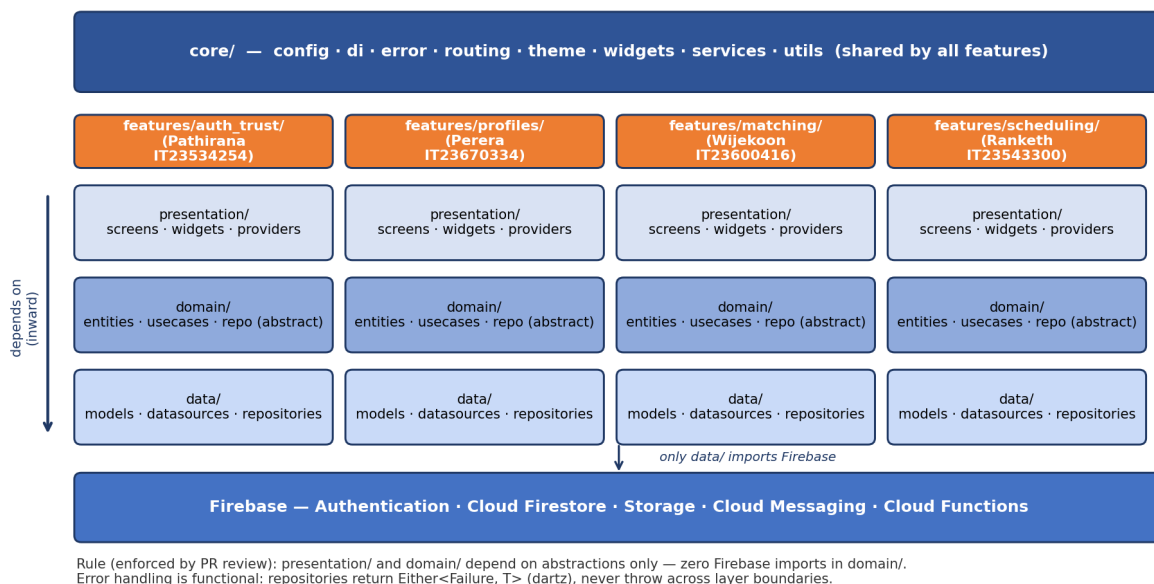


Fig 3 — Feature-first Clean Architecture: 4 owned verticals over a shared core; Firebase isolated to data/.

8. Working Increment Demo

As of 23 Aug 2026 the project is an actively-developed scaffold, reported honestly rather than overstated: every route/screen renders (sign-in/up with a dev-preview bypass, onboarding, profile, discovery, match results, booking shell, feedback form). Genuinely working end-to-end: all of core/ (accessible theme incl. AccessibilityController, go_router with auth guard, shared services) and features/matching's DefaultMatchingStrategy.score(). Remaining use cases are stubs (UnimplementedError('TODO(owner)')) each owner is closing out through Sprint 2's end. The panel walkthrough: register/bypass → onboarding → profile with accessibility on → browse → ranked match → booking shell → (Sprint 2 close-out) confirm session + feedback.

9. Repository Link and Per-Member Commit Summary

Repository: https://github.com/dilmithp/SE3080-Group18_Assignment01

Every PR needs ≥1 teammate review; changes under core/ need ≥2 reviews.

Member	Branch(es)	Commits (23 Aug)	Note
Pathirana	dev, feature/auth-trust, feature/matching-strategies,	44	—

	feature/simplified-accessibility-mode		
Perera	feature/design-system	11	—
Wijekoon	feature/matching	8	—
Ranketh	feature/scheduling	0 (own account)	Action item below

Action item: the scheduling scaffold on feature/scheduling was staged by Pathirana as a reviewable starting point (commit message notes it is not final under the component owner's authorship). Ranketh must commit his own changes before Sprint 2 closes so commit history evidences his individual contribution.

10. Team Collaboration and Communication

Day-to-day coordination runs through the team's Slack workspace (Group 18) with structured channels and a pinned response-time norm. Sprint tracking runs through a live Trello board ("SE3080 G18 - Assignment 01 Sprint Board") mirroring the backlog's epic/story structure, with one card per member per sprint, due dates matching Section 4's plan, and each card checked complete as the work actually finished — code review happens exclusively via GitHub PRs against dev per CONTRIBUTING.md, no direct commits to dev/main, which is what keeps Section 9's commit history usable as real evidence rather than an end-of-sprint dump.

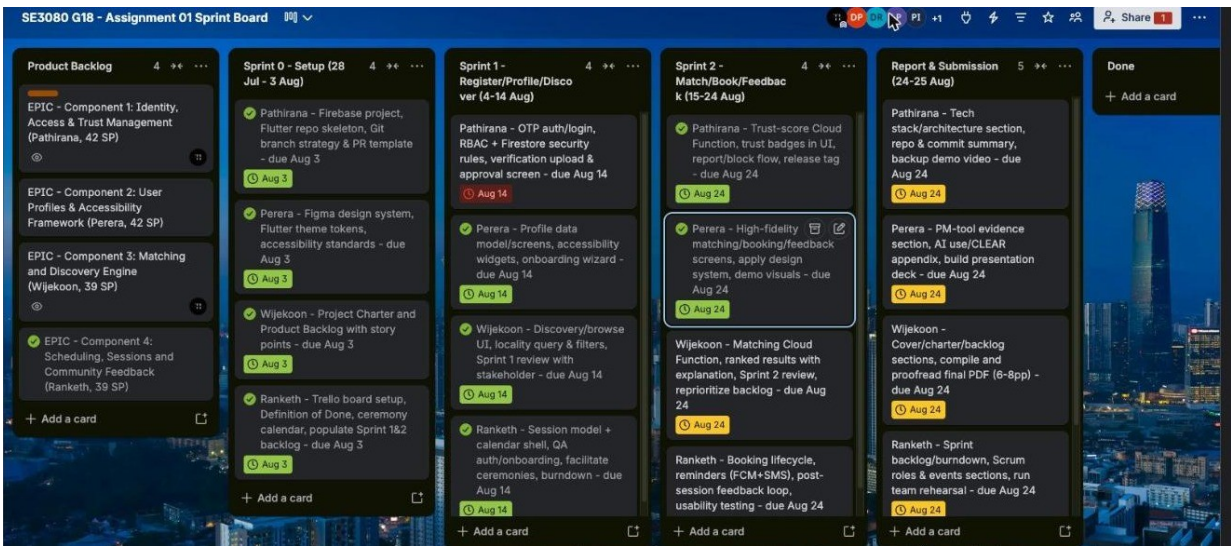


Fig 4 — Live Trello board, captured 23 Aug 2026: Sprint 0 and Sprint 1 fully checked complete; Sprint 2 shows real in-progress state (Pathirana & Perera done, Wijekoon & Ranketh open), matching the Sprint 2 burndown in Fig 2.

Rubric Alignment

Mapped against the Assessment Rubric on the last page of the assignment brief (max 20 marks, scaled from the 100-point breakdown below) so every criterion is easy for the grader to locate.

Criterion	Wt	Addressed in
1. Project Selection and Justification	5	Section 1
2. Agile Project Initiation & Planning	20	Section 2 (Charter) + Section 4 (Sprint 0-2 plans; Sprint 3-4 backlog, Sec. 3.1)
3. Product Backlog & Sprint Execution	20	Section 3 (prioritised backlog) + Section 4 (burndown charts, Fig 1-2)
4. Scrum Roles & Events	25	Section 5 (roles) + Section 6 (ceremony log) + Fig 4 (live Trello board)
5. Team Collaboration & Communication	15	Section 10 (Slack + Trello, Fig 4)
6. Quality of Increment & Demo	10	Section 8 (Working Increment Demo)
7. Submission of Report	5	Full report: all 8 deliverables, repo link, AI-use appendix, PDF, 6-8pp

Appendix A — AI Use Declaration (CLEAR Framework)

Claude (Anthropic) was used to draft and structure this report. All facts — roles, workload weighting, sprint tasks, tech stack, architecture, git history/commit counts — came from the team's own Sprint 0 planning document, the repository's

README/CLAUDE.md/CONTRIBUTING.md/DATA_MODEL.md, and actual git log output. Charts/diagrams were generated from that same data, not invented; Fig 4 is an unedited screenshot of the team's real Trello board, not a mock-up.

- Collaborate: AI structured/drafted prose from the team's existing planning artefacts, not the plan itself.
- Learn: every section maps to a source the team already understood.
- Evaluate: the team must verify Section 6's event summaries and Section 9's commit gap before submission.
- Acknowledge: this appendix declares the use explicitly.
- Reflect: each member can explain, at the viva, their component and what they personally verified.